

is of μ -measure zero. (Here the key observation is that, since $\|\lambda a(\omega) + \nu b(\omega)\| = \|\lambda a(\omega)\| + \|\nu b(\omega)\|$ for a.e. $\omega \in \Omega$, one has $\|\alpha a(\omega) + \beta b(\omega)\| = \alpha\|a(\omega)\| + \beta\|b(\omega)\|$ whenever $\alpha, \beta \geq 0$ for a.e. $\omega \in \Omega$.) Redefining the values of a, b, x, y , and z on H to become 0, the redefined a, b, x, y , and z are the same functions in $L_1(\mu, X)$ as the original ones. The existence of the desired W now follows from $(\#)$. $\square$

The following two lemmata may be known; however, we do not know of any references for them.

Lemma 2.2. *Let μ be a non-zero finite (countably additive non-negative) measure on a σ -algebra Σ of subsets of a set Ω , and let X be a Banach space. Let $z \in S_{L_1(\mu, X)}$, let $E \in \Sigma$, and let $0 < \varepsilon < 1$.*

- (a) *If $z(\omega) \neq 0$ for every $\omega \in E$, then there are $n \in \mathbb{N}$ and pairwise disjoint measurable subsets $E_0, E_1, \dots, E_n$ of E with $\bigcup_{i=0}^n E_i = E$ such that*
- (1) $\mu(E_0) < \varepsilon$;
 - (2) $\left\| \int_{E_i} z \, d\mu \right\| > (1 - \varepsilon) \int_{E_i} \|z\| \, d\mu$ for every $i \in \{1, \dots, n\}$ with $\mu(E_i) > 0$;
 - (3) for every $i \in \{1, \dots, n\}$ with $\mu(E_i) > 0$, there is a neighbourhood W_i of z in the relative weak topology of $B_{L_1(\mu, X)}$ such that

$$(2.1) \quad \left\| \int_{E_i} w \, d\mu \right\| > (1 - \varepsilon) \int_{E_i} \|z\| \, d\mu \quad \text{for every } w \in W_i;$$

this neighbourhood can be chosen to be of the form $W_i := \{w \in B_{L_1(\mu, X)} : \left| \int_{E_i} w_i^(w - z) \, d\mu \right| < \delta_i\}$ for some $w_i^* \in S_{X^*}$ and $\delta_i > 0$.*

- (b) *There is a neighbourhood W of z in the relative weak topology of $B_{L_1(\mu, X)}$ such that*

$$(2.2) \quad \int_E \|z\| \, d\mu - \varepsilon < \int_E \|w\| \, d\mu < \int_E \|z\| \, d\mu + \varepsilon \quad \text{for every } w \in W.$$

Proof. (a). Assume that $z(\omega) \neq 0$ for every $\omega \in E$. Choose a real number $\sigma > 0$ so that $\mu(E \setminus D) < \frac{\varepsilon}{2}$ where $D := \{\omega \in \Omega : \|z(\omega)\| > \sigma\}$. Since the function z is essentially separably-valued, there is a subset N of D such that $\mu(N) = 0$ and the set $z(D \setminus N)$ is separable. It follows that there are pairwise disjoint measurable subsets E_i of E , $i = 1, 2, \dots$, such that $\text{diam } z(E_i) < \frac{\varepsilon\sigma}{2}$ for every $i \in \mathbb{N}$, and $D \setminus N = \bigcup_{i=1}^{\infty} E_i$. Pick an $n \in \mathbb{N}$ so that $\mu(C) < \frac{\varepsilon}{2}$ where $C := \bigcup_{i=n+1}^{\infty} E_i$, and set $E_0 := (E \setminus D) \cup N \cup C$.

Fix an $i \in \{1, \dots, n\}$ and suppose that $\mu(E_i) > 0$. Letting $z_i \in z(E_i)$ be arbitrary and picking a $z_i^* \in S_{X^*}$ so that $\text{Re } z_i^*(z_i) = \|z_i\|$, one has

$$\begin{aligned} \left\| \int_{E_i} z \, d\mu \right\| &\geq \text{Re } z_i^* \left(\int_{E_i} z \, d\mu \right) = \int_{E_i} (\text{Re } z_i^*(z_i) + \text{Re } z_i^*(z - z_i)) \, d\mu \\ &\geq \int_{E_i} (\|z_i\| - \|z - z_i\|) \, d\mu \geq \int_{E_i} (\|z\| - 2\|z - z_i\|) \, d\mu \\ &> \int_{E_i} (\|z\| - \varepsilon\sigma) \, d\mu \geq \int_{E_i} (\|z\| - \varepsilon\|z\|) \, d\mu = (1 - \varepsilon) \int_{E_i} \|z\| \, d\mu. \end{aligned}$$

Now pick a $w_i^* \in S_{X^*}$ satisfying $\text{Re } w_i^* \left(\int_{E_i} z \, d\mu \right) = \left\| \int_{E_i} z \, d\mu \right\|$ and set $W_i := \{w \in B_{L_1(\mu, X)} : \left| \int_{E_i} w_i^*(w - z) \, d\mu \right| < \delta_i\}$ where $\delta_i := \left\| \int_{E_i} z \, d\mu \right\| - (1 - \varepsilon) \int_{E_i} \|z\| \, d\mu > 0$.